



MARKET HUB

Multi-Vendor Marketplace Platform

DEPI GRADUATION PROJECT

ASP.NET Core MVC

Clean Architecture

SQL Server

Entity Framework Core

Project Overview

Market Hub is a web-based e-commerce platform that allows multiple independent vendors to sell products through one centralized application.

The system separates responsibilities across three distinct user roles, and is built on Clean Architecture to keep the codebase scalable, maintainable and secure.



Administrator



Vendor



Customer



Scalability

Grows cleanly as vendors and traffic increase.



Maintainability

Clean Architecture keeps modules decoupled.



Security

Role-based access protects every layer.

Objectives



Scalable Marketplace

Build an online marketplace that grows with demand.



Smooth Shopping

Give customers a smooth, reliable shopping experience.



Secure Access

Provide secure authentication and authorization.



Full Platform Control

Give administrators full control over the platform.



Vendor Autonomy

Allow vendors to manage their own products.

05

Core objectives guiding every design and implementation decision in Market Hub.

HOW IT'S STRUCTURED

System Architecture



Presentation Layer

ASP.NET Core MVC — controllers, views and UI.



Application (Services) Layer

Encapsulates the application's business logic.



Infrastructure (Data) Layer

EF core and external services.



Domain Layer

Core entities and business models.



Dependency Injection

Used across every layer to keep components loosely coupled and easily testable.

Technologies



ASP.NET Core MVC

.NET 10



C#

Programming language



Entity Framework Core

ORM & migrations



SQL Server

Relational database



ASP.NET Core Identity

Auth & user management



Bootstrap 5

HTML5 · CSS3 · JavaScript

Database Design

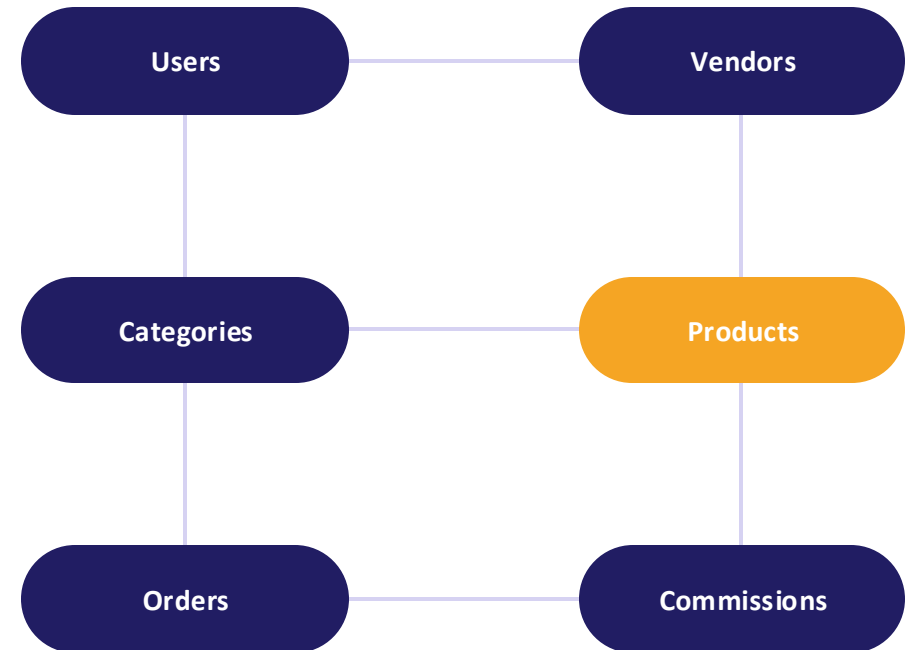
A normalized relational database models the relationships between core entities across the platform.



Relationships between Users, Vendors, Categories, Products, Orders and Commissions.



Entity Framework Core Migrations manage schema updates over time.



Authentication & Authorization



Secure Registration & Login

Implemented using ASP.NET Core Identity.



Role-Based Authorization

Distinct permissions for Admin, Vendor and Customer.



Cookie Authentication

Session state managed through secure cookies.



Protected Actions

Controller actions guarded with Authorize attributes.



Welcome Back

Sign in to your customer or vendor account

Email

you@example.com

Password



Remember me?

[Forgot Password?](#)

Sign In

DEMO ACCESS CREDENTIALS

Admin

Vendor

Customer

New to MarketHub? [Create an account](#)



Create Account

Start purchasing products or list your goods for sale

Email

Password

Confirm password

Register As

Buyer (Customer)



First Name

Jane

Last Name

Doe

Sign Up

Already have an account? [Sign in](#)



ROLE

Admin Module

Full oversight of the marketplace — from categories and vendors to commissions and platform health.



Manage Categories

Create, edit and organize product categories.



Manage Vendors & Customers

Oversee every account on the platform.



Review Products

Approve or reject vendor product submissions.



Monitor Orders & Activity

Track orders and overall platform activity.



Commissions & Reports

Manage commissions and generate reports.



Marketplace Admin Console

Supervising multi-vendor operations, payout transactions, and system diagnostics.

[Manage All Vendors](#)[Pending Products \(0\)](#)

TOTAL MARKETPLACE VOLUME

\$174.95

PLATFORM FEE PROFITS

\$17.50

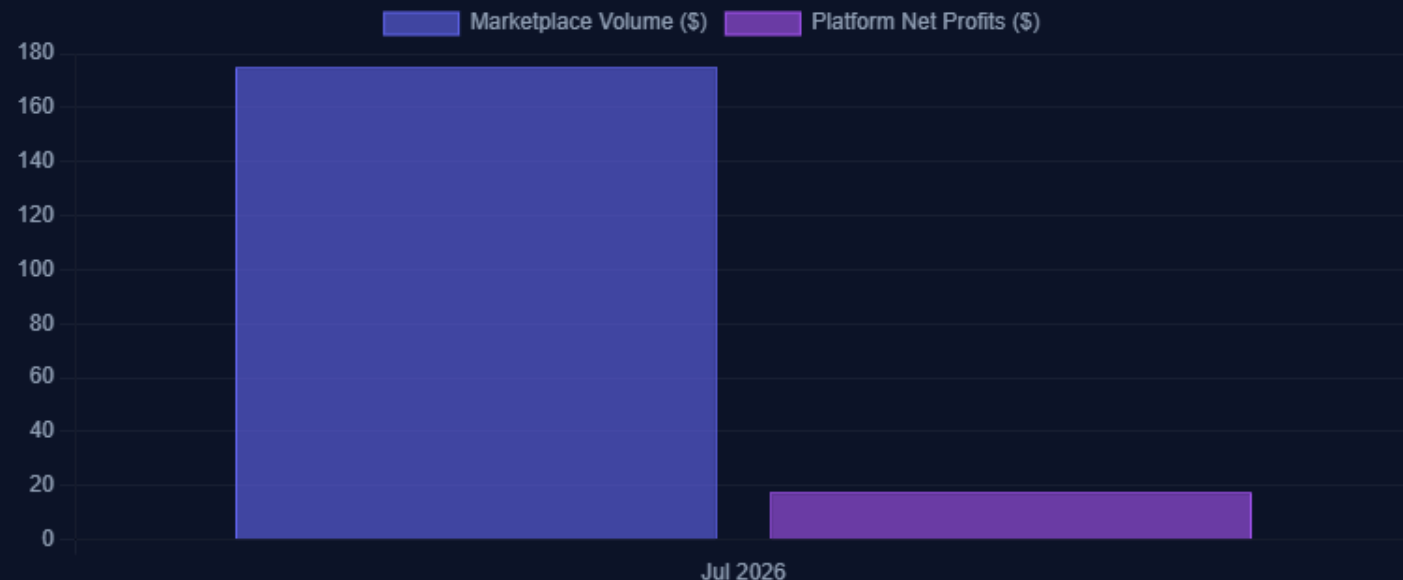
ACTIVE VENDORS

1 (0 Pending)

TOTAL PURCHASES

1

Marketplace Growth & Commissions



Pending Vendor Reviews

No vendor applications requiring approval.

System Status

• Active (99.98%)

DB Engine Connection

SQL Server (Docker)



Marketplace Vendors

Global list of all registered store sellers and payout setups.

[Back to Panel](#)

STORE NAME		OWNER EMAIL	JOINED DATE	COMMISSION FEE	STRIPE ACCOUNT ID	APPROVAL STATE
	Artisanal Wonders Beautifully crafted ceramic and wooden goods made locally.	vendor@marketplace.com	07/10/2026	10%	acct_mock_payout_12345	✓ Live



Pending Product Reviews

Review new submissions from sellers and approve them to list on the public marketplace.



Home Decor

Store: Artisanal Wonders

sports shoes
nice red sports shoes

Price

\$9.99

Approve & Publish to Marketplace



ROLE

Vendor Module

A dedicated workspace for vendors to manage their own storefront, inventory and sales.



Vendor Dashboard

A central view of store performance.



Create, Edit & Delete Products

Full control over the product catalog.



Upload Product Images

Attach visuals to every listing.



Track Orders & Sales

Monitor customer orders in real time.



Vendor Dashboard

Managing Store: Artisanal Wonders



GROSS SALES

\$174.95



NET EARNINGS

\$157.45



MARKETPLACE FEES

\$17.50



TOTAL PRODUCTS

3



Sales Analytics



Recent Sales

- Handcrafted Ceramic Mug** +\$67.47

Qty: 3 • Buyer: Jane Doe Transferred

07/10/2026 07:35 PM
- Carved Wooden Jewelry Box** +\$89.98

Qty: 2 • Buyer: Jane Doe Transferred

07/10/2026 07:35 PM

Manage Product Listings



✔ Product updated successfully.

My Products

Manage your catalog items and check administrator approvals

[Add New Product](#)

PRODUCT DETAILS	CATEGORY	PRICE	STOCK	LISTING STATUS	ACTIONS
 Handcrafted Ceramic Mug Each mug is individually hand-thrown on the wheel. Food, microwave, and dishwasher...	Handmade Crafts	\$24.99	9	✔ Live (Approved)	Edit Delete
 Carved Wooden Jewelry Box Hand-carved walnut jewelry box with velvet-lined interior compartments.	Home Decor	\$49.99	3 (Low)	✔ Live (Approved)	Edit Delete
 sports shoes nice red sports shoes	Home Decor	\$9.99	11	⌚ Pending Admin	Edit Delete

[← Back to Catalog](#)

Add New Product

Specify listing details. The product will require admin approval before listing.

Name

e.g. Handmade Ceramic Pitcher

Description

Detail the materials used, sizes, and care guidelines...

Price

29.99

Stock Level

10

Product Category

Handmade Crafts



Image URL

Leave empty for fallback card image

Publish Listing (Pending Review)

[← Back to Catalog](#)

Edit Product Details

Make adjustments to pricing, inventory, or description.

Name

Handcrafted Ceramic Mug

Description

Each mug is individually hand-thrown on the wheel. Food, microwave, and dishwasher safe.

Price

24.99

Stock Level

12

Product Category

Handmade Crafts



Image URL

https://images.unsplash.com/photo-1

Save Product Changes



ROLE

Customer Module

A smooth, intuitive shopping journey — from browsing to checkout to order history.



Browse Categories & Products

Explore the full marketplace catalog.



Search & Filter Products

Find products quickly with filters.



Shopping Cart & Checkout

A streamlined path to purchase.



View Order History

Track past purchases at any time.



MULTI-VENDOR ECOSYSTEM

Find unique products from creative makers

Discover artisanal crafts, retro apparel, and custom home items sold directly by independent designers. Every transaction supports small vendors globally.

🔍 Search handcrafted mugs, leather jackets...

Search Catalog

Categories

All Categories

Handmade Crafts

Home Decor

Vintage Apparel



Home Decor

Stock: 5

Carved Wooden Jewelry Box

Hand-carved walnut jewelry box with velvet-lined interior compartments.



Handmade Crafts

Stock: 12

Handcrafted Ceramic Mug

Each mug is individually hand-thrown on the wheel. Food, microwave, and...

[← Back to Marketplace](#)

HANDMADE CRAFTS

Added July 10, 2026

Handcrafted Ceramic Mug

\$24.99 USD

DESCRIPTION

Each mug is individually hand-thrown on the wheel. Food, microwave, and dishwasher safe.

**Artisanal Wonders**

Independent Creator • Standard Marketplace Vendor

Quantity

1



✓ In Stock (9 remaining)



Add to Shopping Cart



Shopping Cart

	Handcrafted Ceramic Mug <i>Sold by: Artisanal Wonders</i> \$24.99 each	3	\$74.97 Remove
	Carved Wooden Jewelry Box <i>Sold by: Artisanal Wonders</i> \$49.99 each	2	\$99.98 Remove

Empty Shopping Cart

[Continue Shopping](#) →

Order Summary

Items count	5
Subtotal	\$174.95
Tax / Shipping	FREE

Estimated Total

\$174.95

[Proceed to Checkout](#)

 Secure Checkout via Stripe Sandbox



Delivery details

Shipping Information

Full Name

Jane Doe

Contact Email

jane.doe@example.com

Street Address

123 Creative Lane

City

San Francisco

State / Province

CA

Postal Code

94107

 Check out

Your Order

Handcrafted Ceramic Mug

Qty: 1

\$24.99

Items Total

\$24.99

Shipping fee

FREE

Order Total**\$24.99**



Payment Completed!

Order #1 has been successfully processed

ORDER DATE

July 10, 2026 - 07:35 PM

SHIPPING TO

123 Creative Lane, San Francisco, CA 94107

PAYMENT STATUS

Paid

Purchased Items



Handcrafted Ceramic Mug

Qty: 3 • Price: \$24.99

Sold by: Artisanal Wonders

\$74.97



Carved Wooden Jewelry Box

Qty: 2 • Price: \$49.99

Sold by: Artisanal Wonders

\$99.98

Total Charged

\$174.95

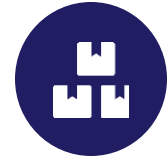
Key Features



Authentication & Authorization



Category Management



Product Management



Shopping Cart



Order Management



Responsive User Interface

WHAT'S NEXT

Future Enhancements



Online Payment Integration

Add Stripe for secure, real-time payments.



Product Reviews & Ratings

Let customers rate and review purchases.



Wishlist

Allow customers to save products for later.



Real-Time Notifications

Keep users updated on orders and activity.



Analytics Dashboard

Give admins deeper insight into performance.



Conclusion



Demonstrates modern ASP.NET Core development practices.



Follows Clean Architecture to improve long-term maintainability.



Built to extend easily with new modules and services.